

Question				Marking details	Marks Available				
					AO1	AO2	AO3	Total	Maths Prac
5	(a)	(i)		High frequency / high energy / short wavelength (1) electromagnetic wave (1)	2			2	
		(ii)		Easily pass through the body / highly penetrative(1) with low ionising power (1)	2			2	
		(iii)		Thallium–(201) (1) because it will <u>decay quicker</u> than the others / won't be exposed to radiation for too long(1) Do not accept shortest half life		1	1	2	
	(b)			$1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32}(1)$ = 5 half lives (1) time = 5 (ecf) $\times$ 17 (ecf) (1) ans = 85 days (1) only allow ecf if first line arrives at an answer other than 5 and is shown		4		4	3
				Question 5 total	4	5	1	10	3 0